Practice Exam - 1

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A 29-year-old woman with a history of NYHA class 2 chronic systolic heart failure, diabetes mellitus, and hypertension is seen in heart failure clinic. Nine months ago, she was diagnosed in the office with asymptomatic rapid atrial fibrillation. She did not tolerate dose escalation of beta-blockers and was referred for pulmonary vein isolation one month after diagnosis. She was lost to follow up due to social stressors. On returning, she explains that she is now 5 months pregnant. Current medications include sacubitril-valsartan , metoprolol, spironolactone, amiodarone, warfarin and furosemide. Interrogation of her ICD shows no breakthrough atrial fibrillation since her PVI. Her renal function, liver function, INR, and thyroid function tests are normal.

Which pharmacologic agents should be discontinued at this time?

A. metoprolol, furosemide and warfarin

B. sacubitril-valsartan, spironolactone and warfarin

C. sacubitril-valsartan, spironolactone and amiodarone

D. warfarin alone

E. furosemide and metroprolol

Commentary References

Testing point: The board certified AHFTC specialist should recognize medications that are contraindicated when pregnancy occurs in the setting of heart failure.

Answer: C. sacubitril-valsartan, amiodarone and spironolactone.

The basic principles of medication use in pregnancy and lactation are to determine the necessity, urgency, timing during gestation, and fetal adverse effect of the drug. Because the majority of drugs transfer to the milk, the effects on neonates should be considered. The lowest effective dose should be used. The woman should be counseled on risks and benefits, and provided current data, acknowledging limitations. Women requiring cardiac medication use during pregnancy should be referred to high-risk obstetrical and maternal-fetal medicine specialists for pregnancy management.

Sacubitril-valsartan is not recommended for use during pregnancy. Use of drugs that act on the renin angiotensin system (RAS) during the second and third trimesters induces fetotoxicity (including decreased renal function, oligohydramnios, skull ossification retardation), and increases neonatal complications (renal failure, hypotension, hyperkalemia) and death. There are no controlled data in human pregnancy. During pregnancy, hydralazine and isordil may be substituted for RAS blocking agents.

Amiodarone is contraindicated in pregnancy and should be used only if benefit outweighs risk. Amiodarone use in pregnant women may increase the risk of adverse fetal effects (e.g., neonatal hypo- and hyperthyroidism, neonatal bradycardia, neurodevelopmental abnormalities, preterm birth, and fetal growth restriction). Untreated underlying arrhythmias in pregnant women pose a risk to the mother and fetus. According to the 2018 ESC guidelines, amiodarone should only be used when other therapies have failed. This patient underwent a pulmonary vein isolation and currently has no further ongoing rapid atrial fibrillation. It would be reasonable to stop amiodarone entirely at this time, with or without replacement with an alternative anti-arrhythmic.

Use of spironolactone during pregnancy is considered contraindicated and requires the benefit to be weighed against the risk. In animals, the mineralocorticoid antagonism and antiandrogenic effects of spironolactone canimpair fetal CNS and urogenital formation. There are no adequate and well-controlled studies in humans. Current ESC guidelines as well as expert consensus on this topic recommend against the use of aldosterone inhibition in pregnancy. The patient is well-compensated and should probably not be treated with mineralocorticoid antagonists during the remainder of her pregnancy in the absence of severe worsening of her heart failure.

The other agents are generally considered safe during pregnancy. While furosemide is associated with in utero fetal death in animal models, and there are no controlled data in human pregnancy, there are decades of use in pregnant women without a signal of significant harm. A large meta-analysis showed no significant difference in fetal outcomes in patients using vs not using loop diuretics. Generally benefits are thought to outweigh risks when diuretic therapy is required.

Beta blockers are the most widely used medications in pregnancy and have been the most studied because of the large number of indications. Several large, retrospective studies of hundreds of thousands of treated pregnant women have shown increased risk of preterm birth, intrauterine growth retardation, fetal bradycardia and fetal hypoglycemia, but no increase in serious congenital abnormalities, neural tube defects or oral clefts. Generally, beta-1 selective agents such as metoprolol are preferred to avoid interfering with beta-2 mediated uterine relaxation and peripheral vasodilation.

Warfarin crosses the placenta and is associated with increased rates of embryopathy, miscarriage and stillbirth in a dose-dependent manner, particularly at doses > 5 mg daily. The smallest effective dose should be used and consideration be given to lower INR goals. Warfarin crosses the placenta and carries important fetal risks, particularly during the first trimester. Although warfarin may be continued in select high-risk scenarios, it is often replaced with low-molecular-weight heparin during pregnancy. In this question, however, the medications that are most clearly contraindicated are sacubitril-valsartan, spironolactone, and amiodarone.

CENTRAL ILLUSTRATION Cardiovascular Medications in Pregnancy

Cardiovascular Medications in Pregnancy